



WORLD DEVELOPMENT STATISTICS

Identifying Hollow Growth Countries (1960–2020)

An Exploratory Data Analysis of income, life expectancy, and population — and what happens when economic growth outpaces human development.

COURSE

Data Science

STUDENT

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DATE

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INTRODUCTION & BACKGROUND

Does economic growth always improve lives?

World development statistics track how countries change over time — income, health, and population — so progress can be measured and compared.

Countries are usually judged by GDP or income growth alone. But rising income doesn't always mean people live longer, healthier lives — and rapid population growth can dilute the benefit per person.



Introducing “Hollow Growth”

When income rises sharply but life expectancy barely improves — or population growth outpaces the gains — economic progress is “hollow”: visible on paper but not felt in people's lives.

190+

countries tracked

60

years of history (1960–2020)

3

indicators combined: income, life expectancy, population



PROBLEM STATEMENT

Growth can hide a deeper problem.

Economic growth does not always improve people's quality of life. Some countries show a specific, risky combination of signals:



High GNI growth

Income rises sharply on paper.



Slow life-expectancy gains

Health outcomes barely keep pace.



Rapid population growth

Gains get spread thinner per person.

Together, this pattern creates “Hollow Growth.”

PROJECT QUESTION

“How can we identify Hollow Growth countries using historical development data from 1960–2020?”



Three datasets, one clean story

DATASET INFORMATION

Source	Gapminder Foundation
Time period used	1960 – 2020
Coverage	190+ countries
Datasets	GNI per capita · Population · Life expectancy

RAW FORMAT (BEFORE CLEANING)

country	1960	1961	1962	...
Afghanistan	469	463	460	...
Albania	736	812	845	...
Argentina	4740	4695	4832	...

Population values used shorthand text, e.g. “3.28M”, “400k” — one column per year (wide format).

Shape before cleaning: GNI (191 rows × 252 cols) • Life expectancy (195 rows × 302 cols)
• Population (197 rows × 302 cols)

DATA CLEANING PROCESS

- 1 Raw Data**
3 separate wide-format CSVs from Gapminder
- 2 Reshape**
Melted wide → long format; renamed columns (year, GNI/life_expectancy/population)
- 3 Clean Types**
Converted year & values to numeric; parsed “K/M” shorthand in population
- 4 Merge**
Outer-joined all 3 datasets on country + year
- 5 Handle Missing**
Investigated missingness (bar/matrix/heatmap); dropped incomplete rows
- 6 Final Dataset**
8,892 complete country-year records · 5 columns · 1960–2020



AGENDA

Roadmap for this presentation

01



Data Overview

02



Data Cleaning

03



Exploratory Data Analysis

04



Key Insights

05



Country Classification

06



Findings

07



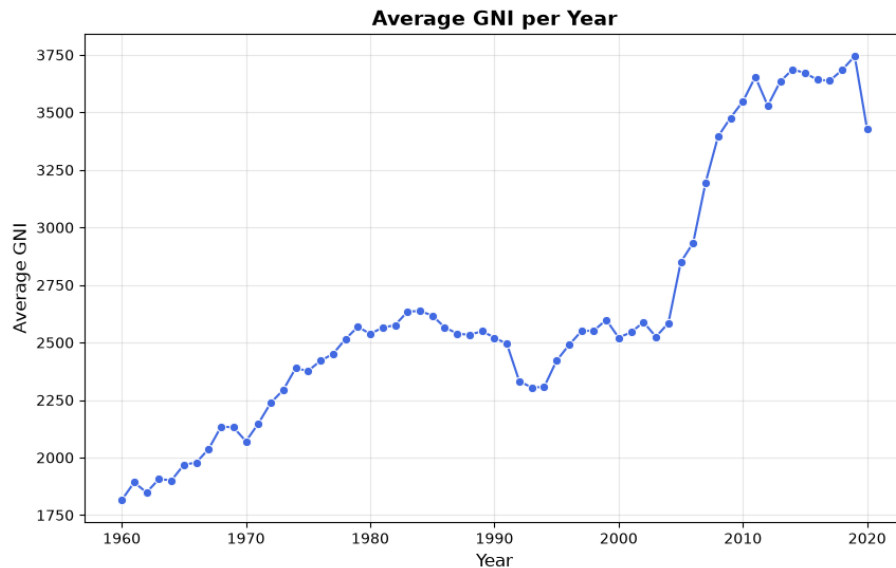
Recommendations



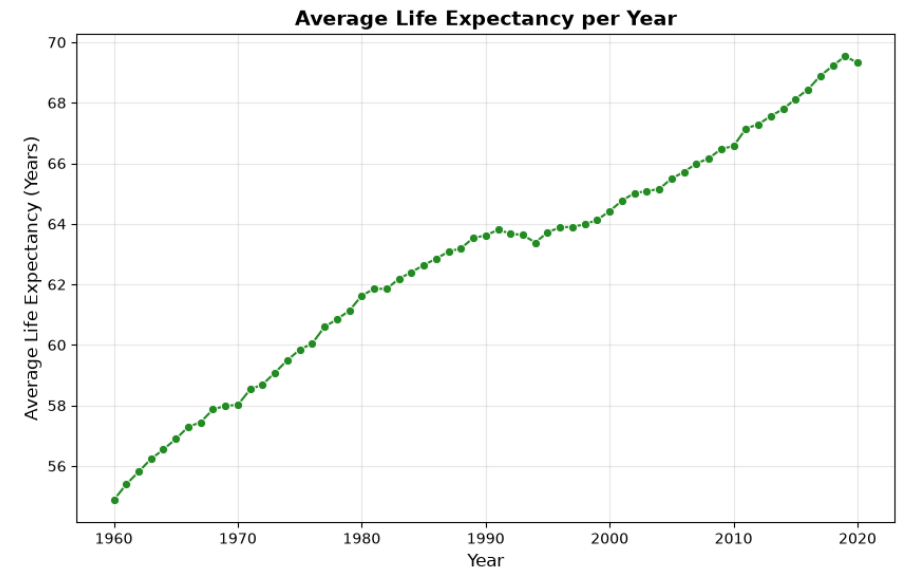
Did the world get richer and healthier together?

Q Did economic growth and life expectancy improve together worldwide, 1960–2020?

Average GNI per Year



Average Life Expectancy per Year



INSIGHT

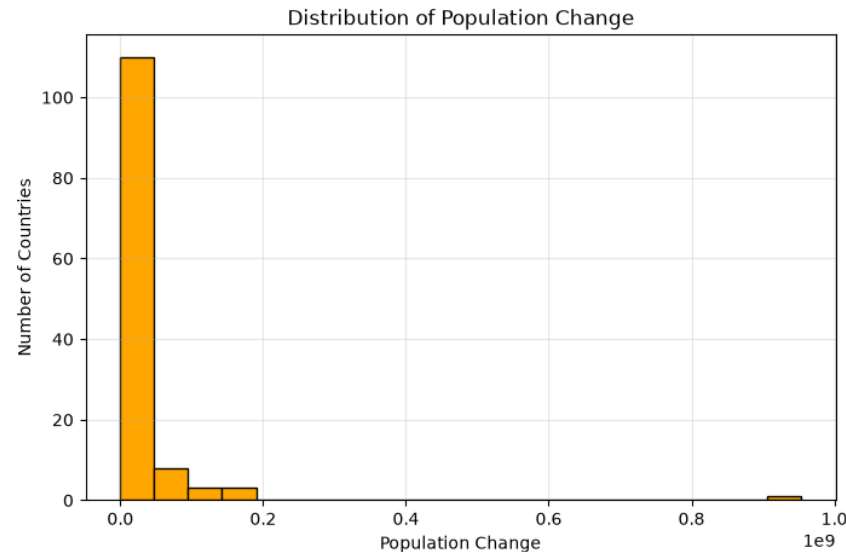
Both global averages rose steadily from 1960 to 2020 — income climbed more sharply after the mid-2000s, and life expectancy improved consistently throughout. But a global average hides very different stories at the country level, which is exactly why we needed to look at each country individually rather than the world average.



How much did population grow across countries?

Q Did population grow at a similar pace across countries between 1960 and 2020?

Distribution of Population Change (1960–2020)



INSIGHT

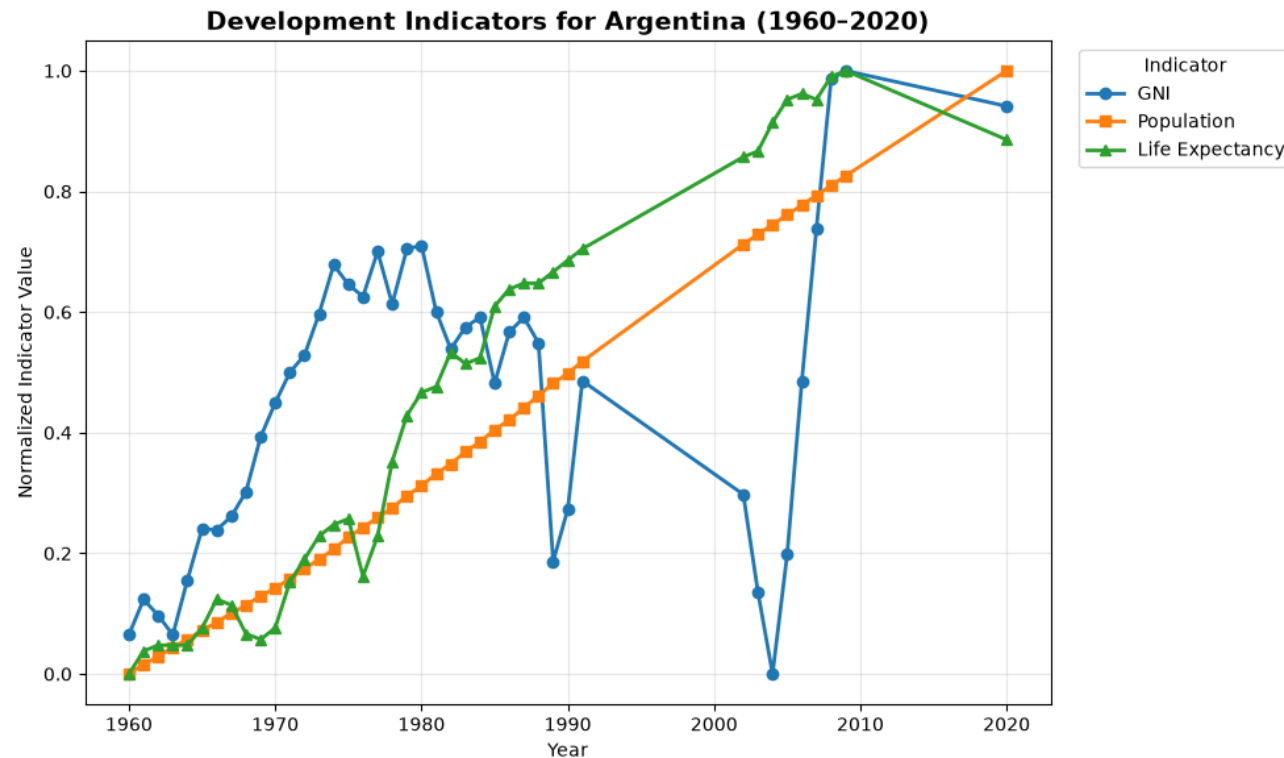
Population growth was highly right-skewed: most of the 125 countries analyzed saw moderate, steady population growth, while a small number recorded exceptionally large increases. When population surges outpace income and health gains, the benefit of growth is spread across far more people — a key ingredient of Hollow Growth.



What does Hollow Growth look like for one country?

Q Argentina shows strong income growth — did life expectancy keep up?

Argentina — Normalized GNI, Population & Life Expectancy (1960–2020)



Population

Increased steadily across the full period.

GNI

Generally increased, but fluctuated year to year.

Life expectancy

Improved more slowly than the other two indicators.

WHY IT MATTERS

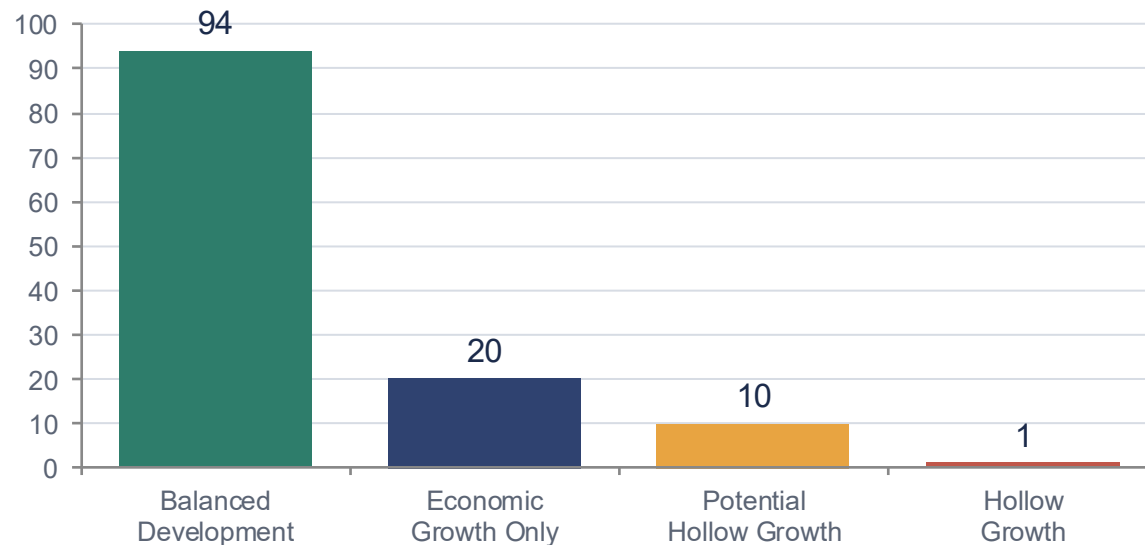
Strong economic and population growth, not matched by comparable health gains, is exactly the imbalance that defines Hollow Growth.



How were countries classified?

Q Using data-driven thresholds, how many countries fall into each development pattern?

Countries by Development Category (n = 125)



Balanced Development

Income and life expectancy improved together.

Economic Growth Only

Strong income growth; health kept a normal pace.

Potential Hollow Growth

Strong income growth; health improved unusually little.

Hollow Growth

Strong income growth; weak health gains; fast population growth.

INSIGHT

Each country was classified using data-driven quartile thresholds (top/bottom 25%) on GNI change, life-expectancy change, and population change — not arbitrary cutoffs. 94 of 125 countries show Balanced Development; only Argentina meets every criterion for full Hollow Growth.



SUMMARY

125 countries analyzed. One clear outlier.

125

countries analyzed

94

Balanced Development

20

Economic Growth Only

10

Potential Hollow Growth



Argentina is the only country meeting every Hollow Growth criterion: high income growth, weak life-expectancy gains, and fast population growth — all at once.

THE OVERALL STORY

- Global averages for income and life expectancy both rose steadily from 1960 to 2020.
- Most countries — 94 of 125 — achieved Balanced Development, with income and health improving together.
- A meaningful minority did not: 30 countries show some degree of unbalanced growth.
- A simple, data-driven quartile rule reliably separates these patterns without arbitrary cutoffs.



RECOMMENDATION

Turning insight into funding action

1



Prioritize funding

Direct development funding first to Argentina, the confirmed Hollow Growth case.

2



Watch closely

Monitor the 10 Potential Hollow Growth countries before they become full cases.

3



Pair money with health

Combine economic investment with healthcare and public-health programs.

4



Keep tracking

Review development indicators regularly, not just once every 60 years.